Hospital Chatbot for Color-Coded Clinical Pathways

Using Bidirectional Encoder Representations from Transformers (BERT)

by

Samuel Quaigraine - 20896511

Samuel Twum - 20885841

Supervisor: Professor Peter Amoako-Yirenkyi

Department of Mathematics
Kwame Nkrumah University of Science and Technology

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Outline of Presentation

Hospital Overcrowding

Source: Citi Newsroom, January 2022 (https://www.citinewsroom.com/2022/01/tamale-teaching-hospital-women-with-newborns-lie-on-the-floor-over-lack-of-beds/)

Problem Statement

Flow management failure

Overcrowding is not only “too many people”: it is a **broken routing system**. Urgency colour does not travel with the patient.

- **Colour left behind.** SATS colour assigned at intake often stops guiding the patient after the first desk.
- **Green congestion.** Non-urgent cases occupy acute capacity and delay Red and Orange care.
- **Text before vitals.** Patients arrive with words; TEWS cannot run until measurements exist.

Research Questions

1. Can the model turn a patient’s complaint text into the right urgency level (1–5)?
2. Can a chatbot give colour and pathway guidance from text before vitals are taken?
3. Can the system refuse non-medical messages before running the model?

Objectives

1. Build a BioBERT classifier that predicts urgency from one English chief complaint.
2. Deploy a chatbot that returns colour and pathway from text at intake.
3. Add a clinical gate so non-clinical chat gets no colour.

Methodology: Predict urgency from text

- **Tokenisation.** Split the English chief complaint into tokens (small pieces of the sentence), and place a special `[CLS]` token first.
- Each token becomes a vector by adding meaning, position, and a segment marker (Lee et al., 2020):

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0). \quad (1)$$

- Attention is a weighted sum. Each token builds Q (what it looks for), K (the label it shows), and V (the meaning it passes on) (Vaswani et al., 2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad d_k = 64. \quad (2)$$

d_k is the length of each Key vector. The `[CLS]` vector at the end is the sentence summary used next.

Methodology: Colour and pathway at intake

- The `[CLS]` summary is scored into five logits $\mathbf{z}$, the raw scores before softmax, one per urgency level:
$$\mathbf{z} = W h_{[\mathrm{CLS}]} + \mathbf{b}. \tag{3}$$
$\mathbf{z}$ is not a probability. A larger z_c means that urgency level looks stronger.
- Softmax turns those scores into five probabilities that add to 1. The largest probability is the urgency level, because the patient gets one colour:
$$\hat{y}_c = \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}. \tag{4}$$
- A fixed map names the colour: $1 \rightarrow$Red, $2 \rightarrow$Orange, $3 \rightarrow$Yellow, 4 or $5 \rightarrow$Green (South African Triage Group, 2012).

Colour and pathway

Colour	Pathway
Red	Resuscitation bay, immediate
Orange	Acute / high-dependency bed, within 10 minutes
Yellow	ED waiting / urgent stream, within 60 minutes
Green	OPD / Minors / Polyclinic, within 4 hours

Levels 4 and 5 both map to Green.

Methodology: No colour for non-clinical chat

Before BioBERT runs, a gate scores how clinical the message is.

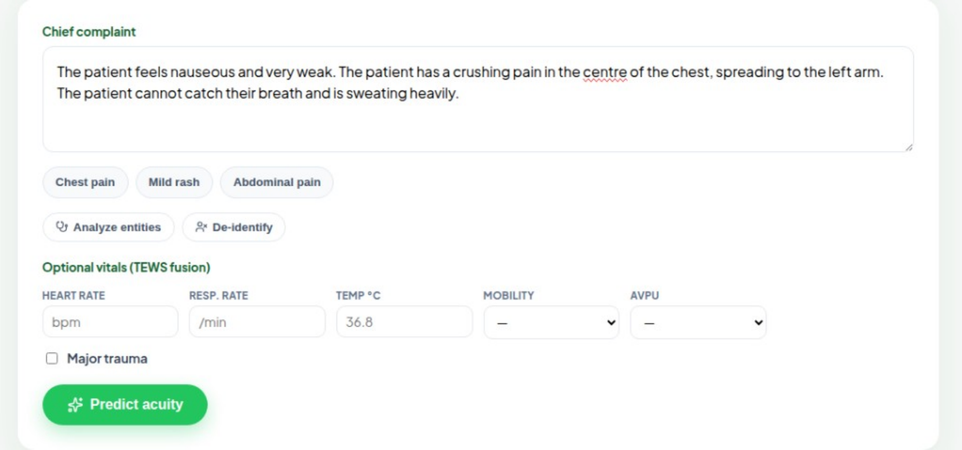
- Symptom words raise the score. Non-clinical chat lowers it:

$$g(X) = \mathrm{clip}\Big(\sum_j s_j(X) - p(X), 0, 1\Big). \tag{5}$$

  $s_j(X)$ is the score for symptom word j. $p(X)$ is the penalty if the message is non-clinical chat. clip keeps the result between 0 and 1: below 0 becomes 0, above 1 becomes 1.

- A colour is given only if $g(X) \geq 0.35$. Otherwise the system asks for a real complaint and stores no colour. This keeps non-clinical chat out of the classifier (Stewart et al., 2023).

Chatbot response: input



Chief complaint entered

Chatbot response: result

Result | Pipeline | Math

Level 3

Yellow

Urgent — physician review needed

Yellow protocol: majors area; standing orders while waiting. Target: Within 60 minutes.

NLP confidence: **99.9985%** · NLP level 3

This is decision support only. Seek clarification from a health professional. | Text-only prediction. Enter optional vitals above to fuse with TEWS.

LEVEL	SATS	MEANING	PROBABILITY
Level 3	Yellow	Urgent — physician review needed	99.9985%
Level 2	Orange	Very urgent — high dependency	0.0012%
Level 5	Green	Routine — lowest clinical priority	0.0001%
Level 1	Red	Life-threatening — immediate resuscitation	0.0001%
Level 4	Green	Non-urgent — stable, can wait	0.0001%

OpenMed entities

Colour and pathway returned

Model Performance

Holdout evaluation

Stratified 10% holdout, $N = 8{,}000$ English Triagegeist complaints (Laitinen-Fredriksson Foundation, 2025).

Metric (baseline BioBERT)	Value
Accuracy	99.92%
Macro-F1	0.9977
Recall Level 1 (Red)	98.14%
Mean latency	$\approx$102 ms / sample

- Strong agreement with nurse acuity labels on held-out English complaints.
- Fast enough for prototype intake assistance.

Impact on Hospital Intake

Traditional intake

Patient arrives → Long queue → Colour assigned late → Destination unclear

Wait: minutes to hours (Mitchell et al., 2023; KATH Emergency Department, 2018)

Chatbot-assisted intake (anticipated)

Complaint entered → Classify ($\approx$102 ms) → Colour + pathway card

Earlier urgency signal; nurse TEWS in parallel (Stewart et al., 2023; South African Triage Group, 2012)

- Clearer next steps for urgent vs routine streams; staff remain in control (Sundström et al., 2014; Nurchis et al., 2025).

Conclusion

1. A BioBERT classifier predicts urgency from one English chief complaint (99.92% holdout accuracy) (Lee et al., 2020; Laitinen-Fredriksson Foundation, 2025).
2. The chatbot returns a SATS colour and a pathway from text at intake, before vitals (South African Triage Group, 2012; Stewart et al., 2023).
3. A clinical gate gives no colour to non-clinical chat (Stewart et al., 2023).

References

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Thank you

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